Piping Material in Refinery

- By Parthasarathi Bandyopadhyay

VARIOUS POINTS THOSE WILL BE COVERED.

- 1)Refinery metallurgy
- 2) Material selection diagram
- 3) Sample P&ID Diagram
- 4) Various pipe classes used in JERP Project.
- 5) Wall thickness as per applicable design code.
- 6) Characteristic logic for piping class nomenclature.
- 7) ASTM Material standard for various piping components say pipe/fittings/valves/bolts made from carbon steel /alloy steel/stainless steel/LTCS/ Monel.
- 8)Special condition for material in JERP Project.
- 9)ASME Dimensional standards for various piping components.
- 10) Discussion on MR0103 applicable to Sour service/Amine service.
- 11)Elevated temperature properties.
- 12) Brittle behavior for material.
- 13)Consideration of various points by Piping Engineer for development of Piping specification.

Refinery metallurgy

An operating superintendent was once asked what his primary responsibility was.

He replied " To keep the hydrocarbons inside the pipes and vessels". Corrosion of process lines and equipment is the major source of leaks in refinery.

We are here because of continuous and varied corrosion.

The choice of materials selected to replace corroded piping or vessel internals depends on the type of corrosions and the availability of suitable material

Sulfur

- High temperature sulfur corrosion is the most common cause of uniform thinning of process equipments. Pitting and crevice corrosion can greatly accelerate failure frequency. High vapor velocity in piping and furnace tubes can produce accelerated corrosion attack due to droplet impingement.
- TP 304 is the most preferred to fight this sulfur corrosion

Pitting

- Often a hydrocarbon leak starts out as a pin hole.
- Small pits in metal have ruined pipe many times. It occurs when the protective surface oxide or sulfide layer is removed in small area.
- Type 316 is recommended to resist pitting because of molybdenum.

Naphthenic acid

- Organic acids present in crude oils are collectively called naphthenic acids. Their corrosive effects on unprotected crude units can be severe .Severe naphthenic acid corrosion occurs primarily in vacuum distillation towers at temperatures in the range of 550 deg to 650 deg F.
- The most cost-effective metallurgy to use in vacuum towers exposed to naphthenic acids attacks is TP 316L

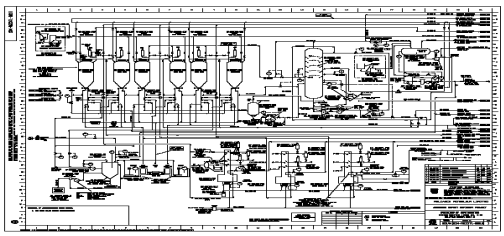
H2S

- Hydrocarbon desulfurizers produce an environments with significant partial pressures of both hydrogen and hydrogen sulfide. This renders the metal more prone to high temperature sulfur corrosion than in the absence of hydrogen.
- Type 304 stainless steel is the preferred alloy for this type of corrosion.

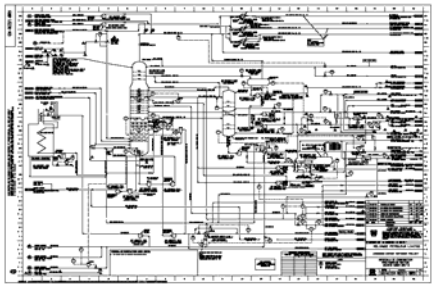
Hydrogen blisters/H2 attack

- Atomic hydrogen, formed from the dissociation of molecular hydrogen may diffuse into steel surface. The hydrogen then recombines into molecular form and becomes trapped inside steel lattice. The subsequent pressure buildup results in blisters that eventually crack the steel. This is common in separator, drums, absorbers and strippers.
- H2 attack of carbon occurs at high temp above 500 deg F. Atomic hydrogen diffuses and decarbonizes steel by combining with carbon to form methane ,which then cause blistering.
- TP 304 is preferred to fight corrosion this type of environment.

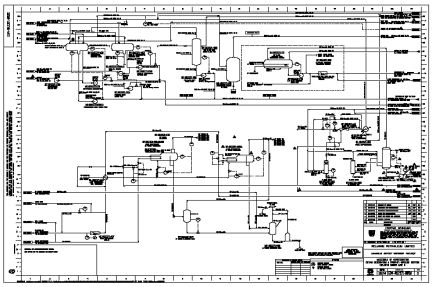
Material Selection diagram- Coking section



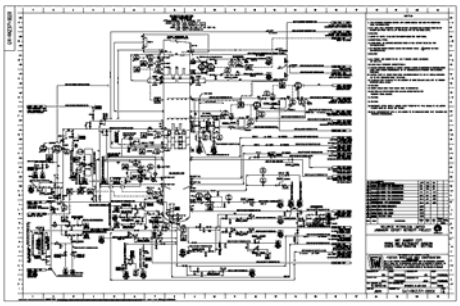
Material selection diagram- Fractionators' section



Steam generation &product cooling



P&ID Diagram for Coker Fractionator bottom



PIPING MATERIAL IN REFINERY

MATERIALS IN ANY PROJECT IS REPRESENTED WITH PIPING CLASS.

REF: 25194-3PS-PL-001 FOR JERP PROJECT

PIPING MATERIAL CLASS- SPECIFY BY SERVICE CLASS ,THE MATERIAL REQUIRED FOR PIPES,FITTINGS,FLANGES & VALVES ETC AND DESIGN DETAILS

FOR BRANCH CONNECTIONS,VENTS,DRAINS,PRESSURE AND TEMPERATURE CONNECTIONS.

VARIOUS PIPECLASS FOR CARBON STEELS ARE

- EC12AB,EC12AC,EC12EA,EC12EB,EC12ED,EC12EG,
- EC12EJ,EC12EK,EC12EM,EC12EN,EC12ER,EC12ES,
- EC12ET,EC12EV,EC12EW,EC12EZ,EC14EA,EC14ED,
- EC14EM,EC14EP,EC14EU,EC14EX,EC16EA,EC16EP,
- EC18EA,EC18ED,EC18EP,EC30EE,EC30IG,EC30IH,
- EC32EA,EC32EB,EC32ED,EC32EF.....EC98EP,
- ECL4EA

VARIOUS PIPECLASS FOR ALLOYSTELLS ARE

- EK14EC,EK14EE,EK16EC,
- EK32EA,EK32EB,EK32EC,EK32EG,EK32EM,
- EK34EM,EK34EA,EK34EB,EK34EC,EK34ED.....
- EK94EA,EK94EC,EK94EE,EK94FE,
- EKL2EB,EKL2EM,EKL2ER

LTCS MATERIAL

- EL12EA,
- EL12ED,
- EL14EA,
- EL32EA,
- EL32ED

NI ALLOY,INCONEL& INCOLLOY

- EN12EM,EN30EM,EN62EA
- EN12EM: Application ; oxygen rich gas(O2)
- EN30EM: Application; Blow down service(BA)
- EN62EA: Application: Blow down hydrocarbon with sour water(BE).

HIGH DENSITY POLYTHELENE

- EP109A, Application: sanitary rising main(SR)
- EP109D: Application: Sulphuric acid(VB), sea water (WS)

STAINLESS MATERIAL

- ES10IA,ES12EA,ES12EB,ES12EG,ES12EJ,
- ES32EA,ES32EB,ES32ED,ES32EG,ES32EH,ES32EJ,
- ES62EC,ES92EC,ESJ2EC

GRP

- EW10AA : design code AWWA C950
  - Application : Sea water.

GLASS REINFORCED THERMO RESIN

- EW10AB : Application : Sea water

DESIGN CODE: ASME B31.3/AWWA C950

- The minimum strength design is guided by
- T= t+c
- t= PD/2(SE+PY), where
- S= Allowable stress at design temperature ,
- E= Weld joint factor ,
- P= Design pressure ,
- Y= Coefficient based on ductility.
- c= Corrosion factor ,

CHARACTERISTIC LOGIC

- Ist E= Eastern, P= Polypropylene,
- 2nd Material Selection,

- C=Carbon steel,
- K= Low alloy steel,
- S= Stainless steel,
- L= Low temperature CS,
- G= Galvanized Carbon steel,
- M=Monel,
- N=Ni alloy, Inconel and Incolloy ,
- P= HDPE, kyner lined FRP,
- U=Alloy 20,
- W=GRP/RTRP/Epoxy lined CS & SS Jacket,

3rd CLASS PRESSURE RATING

- 1 is for 150 lb
- 3 is for 300 lb
- 6 is for 600 lb
- 9 is for 900 lb
- J is for 1500 lb
- L is for 2500 lb

4th digit CORROSION ALLOWANCE

- 2 is for 0.0625"(1.59mm) Corr. Allow.
- 4 is for 0.125" (3.18mm)Corr. Allow.
- 6 is for 0.187"(4.75mm) Corr. Allow.
- 8 is for 0.25"(6.35mm) Corr. Allow

5th &6th digit

- 5th End connection (Construction type)
- Construction Type ,E is for BW ,RF and SW etc
- 6th Sequence No, A thru Z, 1thru 9

Definition of Carbon steel

- Carbon steel as per (AISI) is defined as Carbon steel when no minimum content is specified or required for Chromium, cobalt, columbium, molybdenum, nickel, titanium, tungsten, vanadium
- Or
- any other element to be added to obtain a desired alloying effect.
- When the specified minimum for copper does not exceed 0.4%
- Or
- when the maximum content specified for following elements does not exceed as noted Mn 1.65%,Si 0.6% & Cu-0.6%

Classification of steels

- Steels can be classified by a variety of different system
- Composition such as Carbon, low alloy or stainless steel
- Manufacturing process such as open hearth, basic oxygen process
- Finishing method such as hot rolling or cold rolling
- Product form such as bar plate, sheet, strip, tubing or structural
- De oxidation process such as killed, semi-killed, capped or rimmed
- Micro structure such as ferritic, pearlitic and martensitic
- The required strength level as specified in ASTM Standard
- The heat treatment ,such as annealing, quenching and tempering and thermo-mechanical processing
- Quality factor such as forging and commercial quality

Carbon steel grade for pipes/fittings/castings/plates
Carbon content < = 3%

- Piping- A106 Gr B (killed carbon steel) ,
- IS1239/I S3589/API5L-Gr B/A671 CB6013
- Small fittings-CS A105N GALV
- Large fittings- A234 Gr WPB
- Valve Body- A 216 Gr.WCB
- Plates – A515 Gr.60
- Bolts/Nuts-A193Gr.B7/A194Gr.2H: A307Gr.B/A563 Gr.A.
- Use of Carbon steel is restricted up to temp 400 deg c and (-)29 deg c (up to 12.5 thickness).
- Allowable stress for CS material is 20000 psi upto 300 deg F

Definition of Stainless Steel

- Stainless steels are iron based alloys containing at least 10.5% Cr
- They achieve their stainless characteristics through formation of invisible and adherent chromium rich oxide surface films.
- Carbon is normally present in amounts ranging from 0.03% to over 1.0%
- The selection of stainless steel is based on corrosion resistance, fabrication characteristic, mechanical properties in specific temperature range.
- Stainless steel is available in five category: martensitic, ferritic, austenitic, duplex and precipitation hardening stainless steel

Stainless steel grade for
pipes/tubes/fittings/castings/plates

- Pipes: A312 Gr.TP304/TP316/TP321 seamless:
- A358Gr 304/316/321 welded
- Tube: ASTM A 269 TP 316/316L mainly for HPU tubing
- Small fittings: SS A182-F304/F316/F321:
- Large fittings- SS A403-WP316
- Valve Body- A 351CF8M
- Plates – A240Gr.304/316/321
- Bolts/Nuts-A193Gr.B8 CLASS II/A194Gr.8
- Stainless steel pipes find wider application because it can handle corrosive nature of fluids. These pipes are suitable for low temperature services as low as (-) 250 deg c.

Definition of Low alloy steel

- It exhibits mechanical properties superior to plain CS as a result of addition of alloying elements such as Nickel, Chromium and Molybdenum.
- Total alloy content range from 2.07% up to a level just below that of Stainless steels which contain a minimum of 10% Cr.

Low Alloy steel grade for pipes/fittings/castings/plates

- Pipes:
- : A335Gr.P11(1-1/4Cr-1/2Mo)
- :A335Gr.P22(2-1/4Cr-1 Mo)
- :A335Gr.P5(5Cr-1/2Mo)
- :A335Gr.P22(9Cr-1Mo)
- Small fittings :A182Gr.F11(1-1/4Cr-1/2Mo)
- :A182Gr.F22(2-1/4Cr-1/2Mo)
- :A182Gr.F5(5Cr-1/2Mo)
- : A182Gr.F9(9Cr-1Mo)
- Large fittings : A234Gr.WP11(1-1/4Cr-1/2 Mo)
- : A234Gr.WP22(2-1/4Cr-1/2Mo)
- : A234Gr.WP5(5CR-1/2 Mo)
- : A234Gr.WP9(9Cr-1Mo)
- Valve Body- A217Gr.WC6(1 1/4Cr-1/2 Mo)
- : A217Gr.WC9(2 1/4Cr-1/2Mo)
- : A217Gr.WC5(5CR-1/2 Mo)
- : A234Gr.WC12(9Cr-1Mo)
- Plates – : A387Gr.11(1-1/4Cr-1/2 Mo)
- : A387Gr.22(2-1/4Cr-1/2Mo)
- : A387Gr.5(5CR-1/2 Mo)
- : A387Gr.9 WP9(9Cr-1Mo)
- Bolts/Nuts- : A193Gr.B16/A194Gr.4

Alloy steel pipes

- Steel pipes can withstand temperature beyond 400 deg c
- ASTM P11 can be used up to 510 deg c
- ASTM P22 can be used up to 565 deg c

Tensile strength of some common piping material

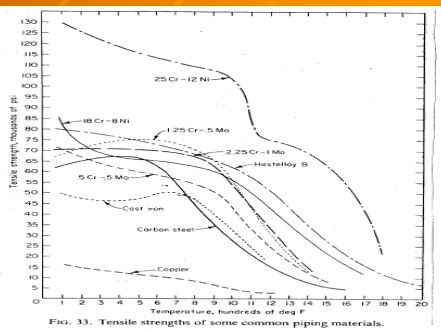


FIG. 33. Tensile strengths of some common piping materials.

Low Temp. Carbon steel grade for pipes/fittings/castings/plates

- Piping- A333 Gr.6
- Small fittings-A350Gr.LF2 Class 1&2
- Large fittings- A420 Gr.WPL-6
- Valve Body- A 352Gr.LCB
- Plates – A516 Gr.60
- Bolts/Nuts-A320 Gr L7.

Monel /NI / Incoloy/HDPE grade for pipes/fittings/castings/plates

- Monel
- Piping : B165/B72
- Large fittings: B366
- Small fittings: B564
- Valve body:
- A494 M35-1
- HDPE
- Piping : IS 4984
- Large fitting: IS4589
- Ni Alloy
- Piping: NI AC B407
- Large fittings:
- NI AL B564
- Incoloy
- Piping: ASTM B423
- Large fittings- ASTM B366
- Small fittings: ASTM B564
- Valve body: ASTM B564

Few Special Condition for Material in JERP Project as per 25194-3PS-PL-004

- For Pipes
- Made from CS
- Max C content: 0.20%
- Max CEN: 0.35%
- Maxm Mn<=0.70%
- Maxm hardness shall be 237 BHN for seamless pipe A106 gr B
- Impact testing is required for WT>=25.4 mm
- Pipe to ASTM A671 CB/C 60 &ASTM A 672 C70 class 11&12 impact test for weld metal deposit & HAZ with a WT>=25.4mm
- Made from LTCS (ASTM A333Gr6)
- Maxm Mn<=0.70%
- Product analysis content of sulphur
- As per NACE -0175 shall be <0.01
- Made from ASTM A691 gr 1-1/4 class 22 EFW/A335 P11
- Sn<=0.015%
- P<=0.012%, the maximum hardness of HAZ and weld metal after stress relieving shall be 225 BHN.
- For Valves
- Made from CS
- Max C content:0.25%
- Max CEN: 0.43
- Valves in accordance to NACE MR-01-75 any major weld repairs to cast bodies and bonnets shall be post weld heat treated.
- Valves made from LTCS:
- Shall be impact tested at (-)45 deg c,
- Acceptance criteria shall be minimum energy level of 27J on average.
- Shall be suitable for continuous operation from(-)45 deg c to (+)110deg cen
- Made from F11/WC6
- Sn<=0.015%
- P<=0.012%

Few important Codes & standards

- B1.1 Unified Inch Screw threads
- B1.20.1 Screw threads –pipe threads ,general purpose
- B16.1 Cast iron pipe flanges& flanged fittings
- B16.5 Steel pipe flanges
- B16.9 Factory made wrought steel butt-welding fittings
- B16.10 Face to face & end to end dimension of ferrous valves
- B16.11 Forged steel fittings ,socket welding and threaded
- B16.20 Ring joints gaskets &grooves for steel pipe flanges
- B16.21 Non metallic flat ring gaskets for pipe flanges
- B16.25 Butt-welding ends
- B16.34 Steel valves-flanged ,threaded and welding ends
- B16.47 Large diameter steel flanges.
- B16.48 Steel line blanks.
- B31.1 Code for power piping
- B31.3 code for process piping

MR0103

- SCOPE: This standard defines material requirements for resistance to SSC in sour petroleum refining and related processing environments containing H2S either as a gas or dissolved in an aqueous phase with or without the presence of hydrocarbon.
- This standard is directed at the prevention of SSC of equipments and components used in the refinery industry.
- Prevention of SSC in carbon steel materials categorized under :P No-1 in section ix of the ASME boiler and pressure vessel code is addressed by requiring compliance with NACE Standard RP0472.

MR0103

- Environmental condition known to cause SSC are those containing free water (in liquid phase) and:
- a) > 50 ppmw dissolved H2S in the free water or
- b) Free water pH<4 and some dissolved H2S present
- c) Free water pH>7.6 and 20 ppmw dissolved hydrogen cyanide(HCN) in the water and some dissolved H2S present
- d) >0.0003 MPa absolute (0.05 psia) partial pressure H2S in the gas in processes with a gas phase
- d.1- The high pH sour environments differentiate refinery sour environment from the oil & gas production sour environments covered by NACE0175/ISO 15156,because many wet sour streams in production also contain CO2 and hence exhibit lower Ph . Another major difference is that chloride ion concentrations tend to be significantly lower in refinery sour services than in oil production sour services.

Factors contributing to SSC

- SSC is defined as cracking of a metal under the combined action of tensile stress and corrosion in the presence of water and H2S. SSC is a form of hydrogen stress cracking resulting from absorption of atomic hydrogen that is produced by the sulfide corrosion reaction on metal surface.
- SSC in refining equipment is affected by complex interaction of parameters including:
- a) Chemical composition , strength

Requirement of CS exposed to aqueous H2S containing 50PPMW or greater

- 1) Be Fully killed
- 2) Have a maximum weld metal macro hardness of 200 HB (Otherwise PWHT required).
- 3) Be seamless(If constructed of welded plate PWHT is required)
- 4) Have a minimum specified tensile strength not exceeding to 4218 kg/cm2
- 5) Utilize valves with 316 SS trim
- 6) SA/A-106GRB meets the above requirement.

Additional condition of CS for Amine exposure

- 1) Be seamless
- 2) Be post weld heat treated at 61 deg cen +-14 deg cen

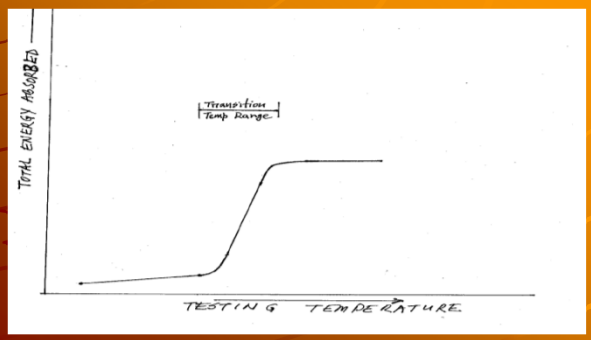
Reason for service failure

- 1. Design,- Structural flexibility ,design notches, joint location of weld joint
- 2.Material- Selection and handling of base &welding materials
- 3.base –metal defects-Introduced during ,
- 4. Fabrication- fabrication, welding, heat treatment
- 5. Service-severe service condition.

Brittle Behavior of metals

- The three conditions which control tendency for steel to behave in brittle fashion include
- 1) High stress concentration
- 2) A high rate of straining
- 3) A low environmental temperature

- Transition temperature for any steel is the temperature above which the steel behaves in a predominantly ductile manner and below which it behaves in a predominantly brittle manner.
- It follows that a steel with low transition temperature is more likely to behave in a ductile manner.
- Metallurgical practice such as de oxidation practice, chemical composition ,rolling ,forging or extruding practice and subsequent heat treatment influence transition temperature of steel .
- In carbon steel piping ,under best condition it can go below (-) 200 deg F.

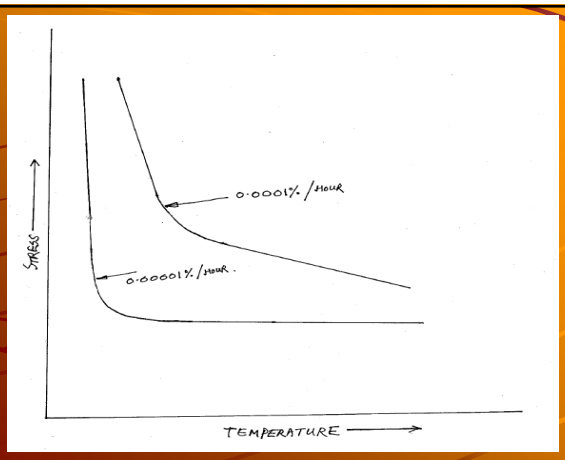


Piping material for Elevated temperature

- **Consideration must be given to the principal mechanical & physical properties.**
- **1) Mechanical properties of importance are**
- **a) tensile strength,**
- **b) proportional limit,**
- **c) Thermal fatigue or shock resistance,**
- **d) Mechanical –fatigue resistance,**
- **e) Torsional elastic limit ,toughness or wearing qualities,**
- **f) toughness or wearing qualities and resistance to wearing,**
- **2) Ability of the material to withstand continued load over periods of long duration without distortion or undue plastic flow. This involve consideration of Creep and stress rupture**
- **3) Surface oxidation, structural changes.**
- **Creep and stress rupture are very important factors**

What is creep & stress rupture ?

- Creep : defined as time dependent part of deformation which accompanies the application of constant load to a solid material .
- Expressed as deformation rate such as 1% in 100,000 hrs.
- Stress rupture: Is the time required for fracture under constant load or more often the stress required to produce fracture in a specified time.
- Hot tensile test data are useful in providing data on ductility/strength/determining the effects of composition on elevated temperature properties.



Why creep test ?

- When a load is applied to a metal at an elevated temperature over a prolonged period of time ,the metal may undergo continuous plastic deformation ,that is it may undergo continuous plastic deformation. This gradual deformation or flow depends on the composition ,the processing and heat treatment of material ,the temperature ,the shape of the section ,the initially applied stress and the time at temperature.
- Thus in the design of piping, it is necessary to know the effects of the three independent variables-stress, time & temperature-on the plastic properties and fracture strength of the materials from which piping is to be constructed.

Factors which affect creep strength

- **The creep & stress to rupture properties are influenced by the metallurgical characteristics of the material and the testing and service environments to which it is exposed.**
- **The most important metallurgical characteristics of the material are**
- **a) Chemical composition. Composition is the most obvious and important variables which affects creep & stress rupture properties. Mere addition of alloying elements does not insure higher creep and stress to rupture properties.**
- **b) Grain size & structure.**
- **Effects of heat treatment : Very important primarily because of their effect on structure**
- **Heat treating temperature and technique which produce a larger grain size generally will improve the creep and stress to rupture properties of steels at elevated temperature.**

Effect of alloy addition on creep strength

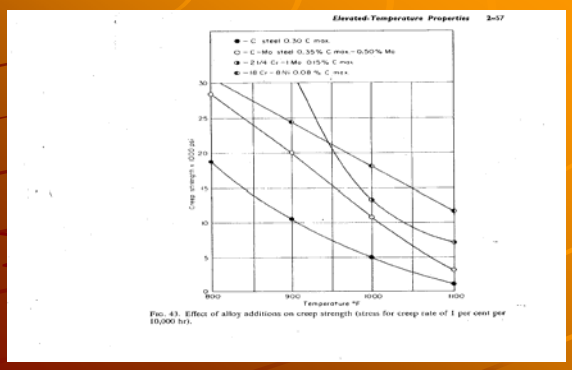


FIG. 43. Effect of alloy additions on creep strength (stress for creep rate of 1 per cent per 10,000 hr).

Effect of alloy addition on rupture strength

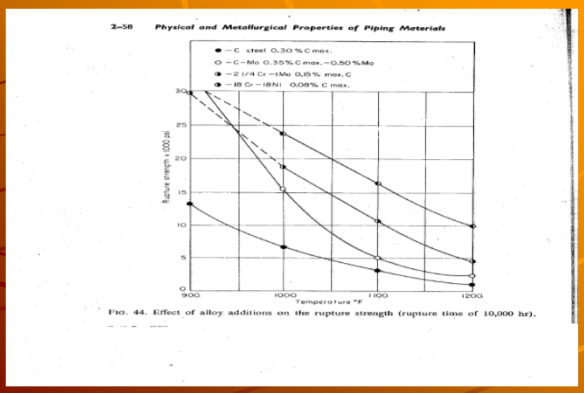


FIG. 44. Effect of alloy additions on the rupture strength (rupture time of 10,000 hr).

Lesson learnt

- 1) Vendor like (Audco /Babcock&Wilcox/ Chero) offered low spec components for valves those got detected during Positive material identification (PMI).
- 2) Piping components made of CS is used even in NACE service without verification of hardness (maxm: 200BHN) in site.
- 3) In some low temp service piping components made of CS material is used leading to various calculation (stress) and test to qualify acceptance of the same.
- 4) Short bolt length as no verification done for the length for blinds for rating above 600# as per ASME B 16.48
- 5) Wrong casting in ball valves supplied by M/S Virgo creating inadequate bolt length projection
- 6)Weak structural supporting member for strainer required on suction of compressor.
- 7)Body blow holes observed inside the valve body of 24" /600 gate valve.
- 8) Procurement of non standard flange (class A) with 30" control valve

PIPING ENGINEERING IN THE DEVELOPMENT OF PIPING SPECIFICATION

- 1) Piping Engineer to review process packages consisting of PFD's ,P&ID's, Guideline to Piping specification. This will help to understand the No of Process fluids& Utility fluids ,their sizes and MOC covered in basic packages.
- 2)Piping Engineer should group the fluids against each piping class
- Based on MOC, Rating taking into consideration local regulation and also process critical requirements related to piping ,valves, corrosion allowances.

3) Piping Engineer should segregate jacketing and tracing requirement ,plastic, lined, and non-metallic piping and allocate separate piping classes.

4) Process Licensor to furnish/confirm item such as design pressure and temperature against each fluid code ,corrosion allowance ,gasket material ,test fluid and radiography requirement.

- 5) Once the group based Piping spec code is finalized ,then it would be ideal to establish minimum and maximum size range from line designation list to make sure the entire size range for each spec.
- 6) Design thickness of each piping component such as thickness of pipe, mitre branches, flanges is calculated based on worst design condition for each piping components.

7) Branching requirements such as set on branch, with or without reinforcements are used on non-critical CS piping/ducting . For critical CS,SS or alloy steel piping standard components such as equal tee, reducing tee , weldolet , reducing tee with reducer are used.

8) In case of low pressure large size duct ,forged flange is uneconomical ,hence plate flange is used.

9) Welded pipes are used only for non-critical fluids which are neither toxic nor hazardous .

- 10) Seamless pipes are used for all critical fluids which are toxic, hazardous or unsafe and where radiography requirement are stipulated or which are jacketed.
- 11) Cast iron components are usually not acceptable to CCOE/SMPV hence shall be avoided while generating piping.
- .

12) In order to accommodate interface between piping & instrumentation for in line &equipment mounted instruments it may be necessary to include

Special components such as screwed coupling ,reducing flanges etc in piping spec.

13) Type of flange i.e screwed ,slip on,lap joint , weld neck shall be based on spec wise design parameters such as pressure & temperature. Piping engineer shall select flanges on criticality of piping system e.g screwed ,S.O and S.W shall not be used beyond temperature limit specified in code

- 14) Type of flange facing is a function of type of fluid being handled i.e. toxic& hazardous or non critical . Flat face facing is generally used for low pressure /temperature system and also as a matching flange for C.I Components . Raised face ,tongue and groove and male &female type of flange facings are used for toxic and hazardous fluid handled at high pressure and temperature.
- 15) Flange finish is a function of type of gasket material selected for non critical and critical services such as serrated for non metallic gasket ,smooth for metallic gaskets.
- 16) Type of gasket material is a function of nature of fluid handled and its pressure and temperature condition.

17) Gaskets such as Spiral wound ,ring joint, graphoil etc are used for critical duty conditions . They are usually used with weld neck flanges having R.F ,M&F,T&G typing facing.
18) Bolting dimension and material standard is also a function of duty conditions and also type of flanges and flange facing .
19) C.S spec is used between the range of (-) 29 deg C and (+) 425 deg C
20) LTCS A.S specs and components are used for temperature below (-) 29 deg C .
21) High temperature A.S specs and components are used for temperature above 425 deg C.
22) SS specs and components can substitute entire range of C.S and A.S specs.